A Branch A cohomological Higgs module: stabilizer-derived quartic overlap and lightness-protection gates

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Abstract

We describe a constrained Branch A cohomological Higgs module, motivated by a Tau Core projection framework, in which the observed four-dimensional Higgs is treated as the visible projection of a tau-profiled parent field. The central reproducible calculation links a minimal $3 + 2$ stabilizer, the canonical hypercharge direction, a Higgs localization exponent $\nu_D = 3/10$, and a zero-mode quartic overlap $I_4(3/10) = 0.133756$. The result is framed as an order-one parent-quartic requirement, not as evidence by numerical coincidence. The construction is a candidate mechanism and validation roadmap. It is not a completed Standard Model derivation, not a proof of Tau Core, and not an empirical claim.

1 Scope and Claim Boundary

This manuscript isolates one theoretical module: a possible origin for a Higgs-scale quartic overlap and a possible cohomological lightness-protection route. The paper does not claim that the Standard Model, gravity, or the full Tau Core parent theory has been derived. The result should be read as a reproducible candidate mechanism with explicit proof obligations.

2 Minimal Projection Setup

The tau coordinate is treated as an internal projection coordinate rather than an ordinary fifth spacetime dimension. A parent Higgs profile is written as

$$H(x^\mu, x), \tag{1}$$

where x^μ is ordinary four-dimensional spacetime and x is the internal tau/projection coordinate. The observed Higgs is the visible projection

$$\phi(x^\mu) = \Pi_{\text{vis}}[H] = \int h_D(x) H(x^\mu, x)\, dx. \tag{2}$$

Projection-null directions have the form $Q_D^\dagger \Lambda$ and are proposed to be quotiented out:

$$H \sim H + Q_D^\dagger \Lambda. \tag{3}$$

3 Branch A Stabilizer

Let the parent internal space split as

$$V \simeq \mathbb{C}^5 = C \oplus W, \qquad \dim C = 3, \quad \dim W = 2. \tag{4}$$

The traceless two-block generator is

$$T_\Sigma = \frac{1}{\sqrt{60}} \operatorname{diag}(2, 2, 2, -3, -3). \tag{5}$$

Its stabilizer is $S(U(3)\times U(2))$, giving the Standard-Model-compatible non-Abelian stabilizer structure. With

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \tag{6}$$

the canonically normalized hypercharge generator satisfies $T_\Sigma = -T_Y$.

4 Higgs Exponent and Zero Mode

The Branch A localization rule is treated here as an assumption/theorem-candidate:

$$\nu_i = \frac{3}{5}|Y_i|. \tag{7}$$

This is the main theoretical blocker of the manuscript. The rule must eventually be derived from a stabilizer-compatible metric, a variational extremum, an index-theoretic constraint, anomaly matching, or another independent principle. Until such a derivation exists, the quartic-overlap result should not be read as a derivation of the Higgs quartic.

The limited claim tested here is narrower. Among linear hypercharge-localization rules $\nu_i = c|Y_i|$, the value $c = 3/5$ is the Branch A working value that keeps the Higgs mode localized and gives an order-one parent quartic requirement. This observation motivates the gate; it does not close it.

For the Higgs doublet, $Y_H = 1/2$, giving

$$\nu_D = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}. \tag{8}$$

The first-order operator is

$$Q_D = \partial_x + \frac{3}{10} \tanh x, \tag{9}$$

and the visible zero mode solves $Q_D h_D = 0$:

$$h_D(x) = \mathcal{N}_D \operatorname{sech}^{3/10} x. \tag{10}$$

5 Quartic Overlap And Sensitivity Audit

For a normalized profile h_ν, the quartic localization functional is

$$I_4(\nu) = \int h_\nu^4 dx = \frac{\Gamma(\nu + 1/2)^2 \Gamma(2\nu)}{\sqrt{\pi}\, \Gamma(\nu)^2 \Gamma(2\nu + 1/2)}. \tag{11}$$

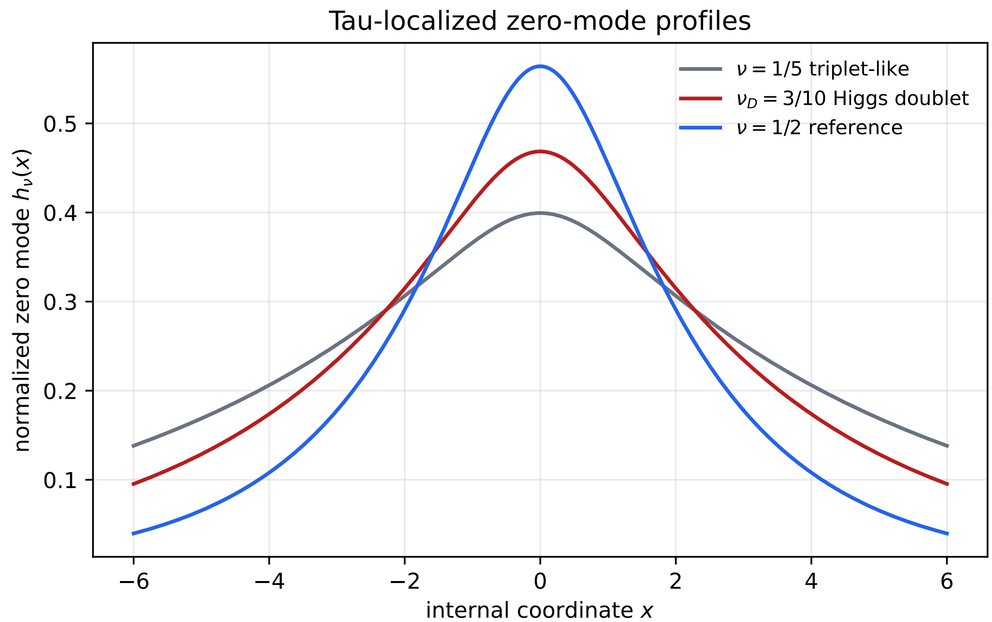

Figure 1: Normalized tau-localized zero-mode profiles. The Higgs doublet exponent $\nu_D = 3/10$ is compared with a triplet-like exponent and a reference exponent.

For $\nu_D = 3/10$ this gives

$$I_4(3/10) = 0.133756. \tag{12}$$

If the parent quartic is $\lambda_\tau(H^\dagger H)^2$, then

$$\lambda_H = \lambda_\tau I_4(3/10). \tag{13}$$

For $\lambda_H \simeq 0.129$, the implied parent quartic is

$$\lambda_\tau \simeq 0.964. \tag{14}$$

The important point is not the exact closeness to one. The safer statement is that the overlap requires an order-one parent quartic rather than an extreme hierarchy. The canonical parent quartic is not derived in this manuscript.

To reduce the numerology risk, the generator writes an explicit sensitivity table. In the moderate window $0.25 \le \nu \le 0.35$, the required parent quartic ranges approximately from 0.847 to 1.129. This means the mechanism is not a delta-function coincidence at $\nu = 0.3$, but it also does not by itself prove the hypercharge-localization rule.

6 Cohomological Protection

The quotient $H \sim H + Q_D^\dagger \Lambda$ suggests a Projection-BRST implementation,

$$sH = Q_D^\dagger c, \qquad sc = 0, \qquad s\bar{c} = b, \qquad sb = 0. \tag{15}$$

At the exact cohomological point the critical operator is factorized,

$$\mathcal{O}_0 = Q_D^\dagger Q_D, \tag{16}$$

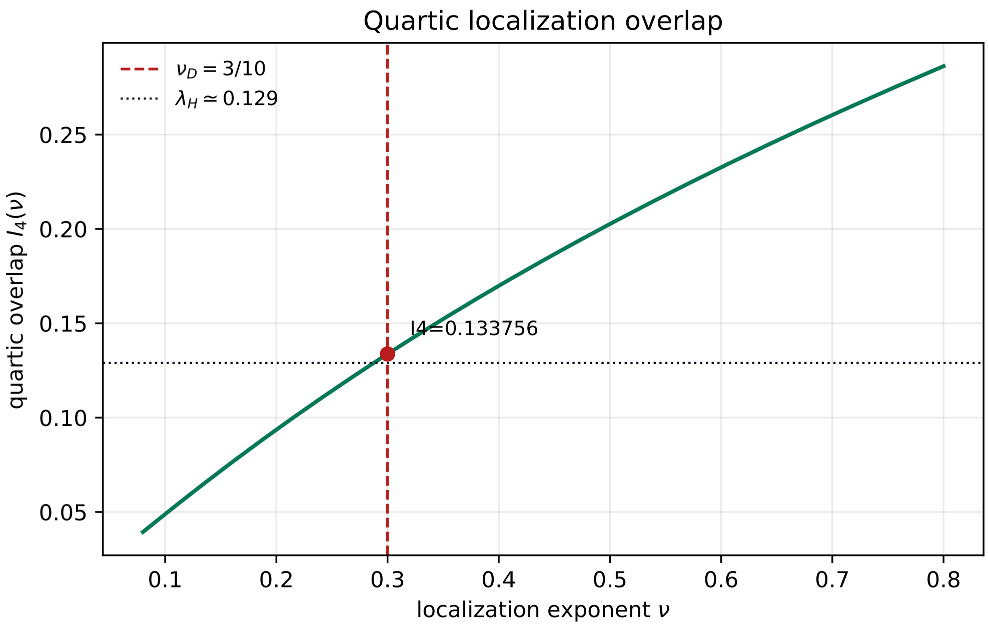

Figure 2: Quartic localization overlap as a function of the tau-localization exponent. The Branch A Higgs value $\nu_D = 3/10$ is marked, but the figure should be read as a sensitivity audit rather than as standalone evidence.

so the visible zero mode is massless. A local parent mass $\int H^\dagger H\, dx$ does not descend to the quotient because it is not invariant under the vertical redundancy. This is only a possible lightness-protection route. The current manuscript does not solve the Higgs hierarchy problem. A complete treatment would require the Hilbert-space domain, nilpotency, regulator, anomaly, and Ward-identity analysis. Here the Projection-BRST language is retained as a roadmap gate, not as a completed proof.

7 Controlled Top/Flavor Deformation

The leading cohomology-breaking deformation is written as

$$\mathcal{O}_D = Q_D^\dagger Q_D - \delta_\star W_D, \qquad W_D(x) = \frac{3}{10}\,\mathrm{sech}^2\, x. \tag{17}$$

The zero-mode expectation value gives

$$\int h_D^2 W_D dx = \frac{9}{80}, \tag{18}$$

and therefore

$$\mu_H^2 = \frac{9}{80}\delta_\star M_\tau^2. \tag{19}$$

This section is a roadmap calculation: the top determinant must still be completed before the deformation can be promoted to a derived prediction. In particular, the manuscript does not yet prove radiative stability. The top/flavor determinant must show that mismatch-independent and linear mass terms are absent or quotient-trivial.

8 What Is Reproduced And What Is Not

The current package reproduces:

- a minimal $3+2$ stabilizer-compatible hypercharge direction;
- the Branch A working exponent $\nu_D = 3/10$ once the $\nu_i = 3|Y_i|/5$ rule is assumed;
- the normalized $\mathrm{sech}^{3/10}$ zero mode;
- the quartic overlap $I_4(3/10) = 0.133756$;
- an order-one parent-quartic requirement and a TeV-scale deformation estimate.

It does not reproduce:

- the full Standard Model;
- the Yukawa hierarchy;
- anomaly cancellation from the parent construction;
- radiative stability;
- the measured Higgs mass from a completed top determinant;
- a collider-ready heavy-sector spectrum.

9 Why This Is Not Just Numerology

The quartic-overlap coincidence alone is not evidence. The module becomes scientifically interesting only if the gates are passed in the right order: first derive the hypercharge-localization rule, then prove the quotient/anomaly safety, then complete the top determinant, and only then compare the resulting heavy-sector predictions to collider constraints. Without those gates, the value $I_4(3/10)$ should be treated as a compact mathematical motivation rather than as physical validation.

10 Near-Term Falsifiable Prediction

The current module predicts a narrow kind of future test rather than an already validated signal. If the top/flavor deformation gate is completed with a natural spurion $\delta_\star \sim 10^-3$, the same equations imply a parent scale M_τ in the multi-TeV range and a Higgs-sector spectral threshold

$$m_{\mathrm{gap}} \simeq \nu_D M_\tau \sim 2\text{–}4\,\mathrm{TeV}. \tag{20}$$

The associated Higgs-coupling deviations should be parametrically of order

$$\frac{\Delta g}{g} \sim \frac{v^2}{m_{\mathrm{gap}}^2} \sim 10^{-3}\text{–}10^{-2}. \tag{21}$$

This is a falsifiable target, not a confirmed prediction. The module weakens if the completed determinant forces much larger deviations, no TeV-sector threshold, or collider-excluded states.

11 Validation Gates

The module would fail if any of the following gates fail:

- the Branch A projection metric rule $\nu_i = 3|Y_i|/5$ cannot be derived;
- the parent quartic is not canonical or requires large tuning;
- the Projection-BRST quotient is anomalous or regulator-dependent;
- a visible Higgs mass is unavoidable at $\delta_\star = 0$;
- the top determinant produces a large mismatch-independent or linear mass term;
- the TeV-sector spectral window is excluded by Higgs coupling or direct-search constraints.

12 Conclusion

The Branch A Higgs module links a $3 + 2$ stabilizer, hypercharge normalization, a $\text{sech}^{3/10}$ visible zero mode, and a quartic overlap requiring an order-one parent quartic. The paper establishes a compact, reproducible candidate mechanism. It does not establish the full parent theory or solve the Higgs hierarchy problem. The next paper-grade step is to derive the $\nu_i = 3|Y_i|/5$ rule, complete the Projection-BRST and top-determinant gates, and compare the implied heavy-sector window against collider constraints.

References